

### **Scenario 1: Identify Consistent High Performers**

**Question:** Find employees who have consistently received a salary increase for the past 3 years.

**Explanation:**

- Perform JOIN on the salary table using employee\_id
- Compare salary data for the past 3 years.
- View the details of employees whose salary is increased.

### **Scenario 2: Customer Retention Analysis**

**Question:** Find customers who made purchases in the last 6 months but not in the previous 6 months.

**Explanation:**

- Use two subqueries to find customers who purchased within specific timeframes using the orders table.
- Display the purchase of the last 6 months.
- Exclude customers who purchased earlier using NOT IN.

### **Scenario 3: Identify Products with Declining Sales**

**Question:** Find products whose sales decreased by more than 30% compared to the previous month.

**Explanation:**

- Use JOIN on the sales table using product\_id.
- Compare sales data of the current month and the previous month.
- Calculate the percentage decrease and filter products with a decline greater than 30%.

### **Scenario 4: Calculate Moving Average of Sales**

**Question:** Calculate a 3-month moving average for product sales.

**Explanation:**

- Calculate the moving average using the AVG() function.
- Partition the data by product\_id.
- Order the data by sales\_date.

### **Scenario 5: Detect Duplicate Records**

**Question:** Find all duplicate records from a customer table based on email and phone number.

**Explanation:**

- Use GROUP BY clause to group records.
- Use HAVING to filter groups with more than one occurrence.

### **Scenario 6: Identify Products with No Sales for 3 Consecutive Months**

**Question:** List products that have not had any sales for 3 consecutive months.

**Explanation:**

- Use a subquery with LEFT JOIN to find missing sales data.
- Use date functions to check for gaps in sales data.
- Identify products with no sales for the required period.